

GEE-LLM: Grounded Google Earth Engine Code Generation and Scientific Interpretation for Geospatial Decision Support

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Highlights

- GEE-LLM closes the complete loop from natural language query to decision-ready geospatial intelligence, combining verified satellite code generation with grounded scientific interpretation.
- A two-pass large language model architecture separates executable code generation from result interpretation, ensuring that the reasoning stage operates exclusively on execution-validated numeric output.
- A local sandboxed execution engine with physical boundary validation raises script correctness from 12–17% (zero-shot baseline) to 44–45% across a 100-query benchmark spanning diverse Indian ecological zones.
- McNemar’s test confirms that improvements are statistically significant ($p < 0.001$), with odds ratios up to 15.0, demonstrating that the self-correction loop recovers failing scripts rather than degrading successful ones.
- The framework operates on commodity CPU hardware without GPU resources or commercial cloud subscriptions, making geospatial decision support accessible to resource-constrained research environments.

Statements and Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Data and Code Availability

The interactive web application is hosted at: <https://geo-ai-llm-aman.streamlit.app/>. The full GEE-LLM software framework, the lexical template corpus, the 100-query benchmark, and all model output logs are publicly available at: <https://github.com/AmanSharma000/gee-llm>.

Authors' Contributions

CRedit taxonomy mapping:

- **Aman Sharma:** Conceptualization, Methodology, Software, Formal Analysis and Investigation, Writing — Original Draft, Writing — Review & Editing.
- **Manish Kumar:** Supervision, Resources, Writing — Review & Editing.

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